StayWise

Hotel Booking Analytics for Revenue & Cancellation Optimization

HOSPITALITY / TRAVEL | SECTION B, GROUP G-1

CORE PROJECT TEAM

Sarthak Mishra (Project Lead)

Ganga Raghuwanshi (Data Lead)

Musthyala Sadhvik (ETL Lead)

Kunal Dev Sahu (Analysis Lead)

Soumen Dass (Visualization Lead)

Deepak Mishra (Strategy Lead)

Lakshay Yadav (PPT & Quality Lead)

Code Repository & Pipeline

github.com/ganga-300/SectionB_G-1_StayWise

Interactive Analytics Dashboards

Executive: .../profile/soumen.dass/viz/StayWise-new1

Operational: .../profile/soumen.dass/viz/StayWise-new2

Context & Problem Statement

WHY HOTEL CANCELLATIONS ARE A REVENUE CRISIS

SECTOR CONTEXT

The hospitality industry loses significant revenue to booking cancellations, exacerbated by frictionless cancellation policies offered by Online Travel Agencies (OTAs). This disrupts staffing, pricing, and RevPAR targets.

TARGET DECISION-MAKERS

Hotel General Managers · Revenue Managers

PROBLEM STATEMENT

What are the primary behavioral and transactional drivers of hotel cancellations, and how can policies be optimized to mitigate this risk?

OBJECTIVE

Identify high-risk booking profiles to deploy targeted deposit policies, direct-channel incentives, and dynamic pricing to maximize net revenue.

41%

City Hotel
Cancellation Rate

27%

Resort Hotel
Cancellation Rate

OTA

Highest-Risk
Booking Channel

Data Engineering: Source to Pipeline

ETL PROCESS — KAGGLE HOTEL BOOKING DATASET

RAW DATA SOURCE

Kaggle Hotel Booking Demand Dataset

July 2015 – August 2017

119,209 rows · 49 columns

→

PROCESSED DATA

Clean, Feature-Rich Analysis Dataset

Python ETL Pipeline

87,219 rows · 29 columns

KEY COLUMNS

`is_canceled` – Target flag

`lead_time` – Days before arrival

`ADR` – Average Daily Rate

`market_segment` – Channel

`deposit_type` – Risk lever

PYTHON ETL PIPELINE · scripts/etl_pipeline.py · notebooks/02_cleaning.ipynb

01 Null Handling

Dropped `company` column (heavily null). Imputed missing values for `country`, `agent`, and `children`.

02 Outlier Treatment

Capped ADR at €5,000 to eliminate erroneous billing entries skewing revenue analysis.

03 Feature Engineering

Created: `total_nights`, `total_revenue`, `clv_proxy`, `is_peak_season` for richer analysis.

KPI & Metrics Framework

FOUR CORE METRICS TIED TO BUSINESS IMPACT · notebooks/03_eda.ipynb

Cancellation Rate %

27.53%

`sum(is_canceled) / count(bookings)`

Direct measure of revenue leakage — the core problem metric. Every policy decision flows from reducing this number.

Total Revenue

€33.9M

`total_nights × ADR`

Gross financial value of all confirmed stays. Underpins dynamic pricing strategy and revenue forecasting.

Revenue per Guest

€197.4

`total_revenue / total_guests`

Measures profitability across demographic segments. Family guests (€180) vs Non-Family (€199) reveal pricing opportunity.

CLV Proxy

Realized Value

`total_revenue × (1 – is_canceled)`

Filters out phantom revenue from cancelled bookings. Shows the true realized economic value per booking.

Key Insights from Exploratory Data Analysis

5 BUSINESS-CRITICAL FINDINGS FROM 87,219 BOOKINGS

01

City Hotel Volatility

41% vs 27%

City Hotels suffer a 41% cancellation rate vs Resort Hotels at 27% — a 14-point gap demanding differentiated policies for each property type.

02

Phantom Demand

> 180 Days

Bookings made more than 180 days in advance cancel at a disproportionately high rate — speculative placeholders, not committed revenue.

03

Deposit Effect

~0% Cancel

"No Deposit" bookings account for nearly all cancellations. Non-Refundable deposits are proven to virtually eliminate cancellation risk.

04

Third-Party Risk

OTA = 35.4%

Online TA/TO channels cancel far more often than direct hotel bookings, compounding commission costs with unpredictable revenue leakage.

05

Peak Season Cost

July–August

Summer peak drives highest ADR (€106.02 avg) and booking volume — making cancellations during this window the most financially damaging.

Advanced Analysis

HYPOTHESIS TESTING · RISK COMPOUND PROFILING · scipy · pandas · seaborn

HYPOTHESIS TESTING

Independent Samples T-Test

H_0 (Null): No significant difference in lead time between cancelled and fulfilled bookings.

✓ **RESULT: H_0 Rejected**

p-value < 0.001 → 99.9% statistical confidence
Cancelled bookings have significantly higher mean lead times.

BUSINESS IMPLICATION

Lead time is a statistically proven, actionable lever — justifying deposit policy mandates for long-lead bookings with hard data confidence.

LIBRARIES USED · pandas · scipy.stats · seaborn · matplotl

RISK COMPOUND ANALYSIS

The Ultimate High-Risk Profile

Data segmentation identified the exact booking profile to target:

- City Hotel
- No Deposit Policy
- TA/TO (OTA) Channel
- Lead Time > 180 Days

= Laser-Focused Policy Target

Tableau Dashboard Walkthrough

DUAL-LAYER ANALYTICS · EXECUTIVE OVERVIEW + OPERATIONAL DRILL-DOWN

EXECUTIVE VIEW

For: CEO / General Manager

Total Revenue · Cancel Rate · Bookings

KPI summary tiles visible at-a-glance

Monthly Revenue Trend

July–August peak clearly visible (€6.97M peak)

ADR by Price Tier

Premium (€4.3M) → Standard → Economy → Budget

Cancellation by Market Segment

Online TA highest at 35.4%, Corporate lowest at 12.1%

.../soumen.dass/viz/StayWise-new1/StayWise-1

OPERATIONAL VIEW

For: Revenue Analyst / Manager

Lead Time vs Cancellation

Cancellation rate rises sharply beyond 180-day lead time

Customer Segment Revenue

Non-Family (€199) vs Family (€180) revenue per guest

Market Segment Filter

Dynamic filter updates Risk Profile Matrix in real-time

Booking Volume Heatmap

Identify under-priced high-demand date clusters instantly

.../soumen.dass/viz/StayWise-new2/StayWise-2

Strategic Recommendations

THREE DATA-DRIVEN ACTIONS TO RECOVER REVENUE

01 Targeted Deposit Policies

INSIGHT:

Lead time > 180 days + No Deposit = Massive Cancellations

ACTION:

Mandate non-refundable deposits for all bookings made more than 180 days in advance.

IMPACT:

Secures early cash flow and converts speculative placeholders into committed, protected revenue.

02 "Book Direct" Loyalty Program

INSIGHT:

Direct bookings rarely cancel and save 15–25% in OTA commissions per booking.

ACTION:

Offer exclusive perks (free breakfast, parking, room upgrades) for direct web or phone bookings only.

IMPACT:

Drops OTA commission savings straight to net profit while simultaneously reducing cancellation rates.

03 Dynamic Peak-Season Pricing

INSIGHT:

July/August = highest volume + highest ADR. Cancellations here are the most costly.

ACTION:

Deploy 30-day non-refundable cancellation windows and rate premiums during peak summer months.

IMPACT:

Protects highest-value inventory from last-minute vacancy and revenue erosion at peak demand.

Impact & Business Value

THE FINANCIAL UPSIDE OF ACTING ON THESE INSIGHTS

10%

Conversion

High-risk No-Deposit long-lead bookings converted to Non-Refundable

Protects millions in top-line room revenue and eliminates last-minute vacancy scramble

5%

Channel Shift

Bookings shifted from OTA to Direct channels

15–25% commission savings drop straight to net profit — zero additional overhead

Cost of Inaction

Ignoring OTA dominance & speculative booking behavior

Guarantees continued RevPAR erosion, unpredictable staffing & margin compression

THE "SO WHAT?"

These are not theoretical gains. Using existing booking data and zero new technology, these three interventions can meaningfully shift cancellation rate, commission burden, and peak-season revenue — within a single quarter of rollout.

Limitations & Next Steps

HONEST CONSTRAINTS · ROADMAP TO REAL-TIME PREDICTION

LIMITATIONS

✕ Pre-Pandemic Scope

Dataset spans 2015–2017 — misses COVID-19 effects, remote work "bleisure" travel, and post-pandemic booking behavior.

✕ No Competitive Data

Lacks competitor pricing, local event calendars, or macroeconomic indicators that influence demand patterns.

✕ Static Retrospective

Insights are backward-looking. No real-time feed means the dashboard reflects history, not the current booking window.

NEXT STEPS

✓ ML: Cancellation Probability Score

Build a Random Forest / XGBoost classifier assigning a real-time cancel probability at checkout — triggering dynamic deposit rules automatically without manual review.

✓ Live Dashboard Integration

Connect Tableau to a live PMS (Property Management System) API feed for real-time KPI tracking, anomaly alerts, and daily revenue optimization.

SUGGESTED ROADMAP

- Q1: Expand dataset (post-COVID era)
- Q2: Build ML cancel predictor
- Q3: Live PMS API integration
- Q4: Automated pricing rule engine

Team & Contributions

SECTION B, GROUP G-1 · NEWTON SCHOOL OF TECHNOLOGY · STAYWISE ANALYTICS

Sarthak Mishra

Project Lead

Report Writing · PPT & Viva Prep

Ganga Raghuwanshi

Data Lead

Dataset Sourcing · Data Strategy

Musthyala Sadhvik

ETL Lead

ETL Pipeline · Data Cleaning

Kunal Dev Sahu

Analysis Lead

EDA & Analysis · Hypothesis Testing

Soumen Dass

Visualization Lead

Tableau Dashboards · KPI Analysis

Deepak Mishra

Strategy Lead

KPI Framework · Recommendations

Lakshay Yadav

Quality Lead

PPT & Viva Prep · Quality Assurance